



Introduction to the Internet of Things

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History & definitions



- The term "Internet of Things" has been around since the early 2000s [1]. This term refers to autonomous computing devices being networked together to perform various tasks.
- The term was coined by Kevin Ashton of the MIT center and was originally in reference to Radio Frequency Identifier (RFID) information being made available on the Internet.
- This information can be sensed, gathered, parsed, and posted to the internet by way of automated and interconnected computing devices.
- The term "Internet of Things" has since grown to encompass far more applications and technologies than the original RFID reference.
- The Internet of Things is not a new concept, technology, or set of products, but is rather a natural evolution of networked computing technology, enabled primarily through affordable processing and connectivity. IoT is an extension of the "ubiquitous computing" concept popularized by Mark Weiser.

Definitions



- Internet of Things (IoT) refers to the network of physical devices connected to the internet, enabling them to collect and exchange data without human intervention.
- IoT involves the interconnection of various objects or "things" embedded with sensors, software, and connectivity capabilities.
- These connected devices can range from everyday objects like smartphones and wearables to specialized equipment in industries, homes, and cities.
- The data collected by IoT devices can be analyzed and used to automate processes, make informed decisions, and improve efficiency and convenience in various domains.

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Components of IoT (1/2)



- Sensors:** IoT devices are equipped with sensors to collect data from the environment. These sensors can measure various parameters such as temperature, humidity, light intensity, motion, pressure, and more.
- Connectivity:** IoT devices rely on different communication technologies to connect to the internet. This can include Wi-Fi, Bluetooth, cellular networks (3G, 4G, or 5G), LPWAN (Low-Power Wide-Area Network), and even satellite connections.
- Data Processing:** Once the data is collected by IoT devices, it needs to be processed and analyzed. This can occur locally on the device itself or be transmitted to cloud servers for more complex analysis and storage.

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Components of IoT (2/2)



- Actuators: IoT devices often have actuators that can perform physical actions based on the data received. These actions can include turning on or off lights, adjusting temperature, activating alarms, controlling motors, and more.
- Backend Infrastructure: IoT systems require backend infrastructure, including servers, databases, and software platforms, to manage and process the collected data, control devices, and enable communication between different components.

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IoT DEVICES

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Devices



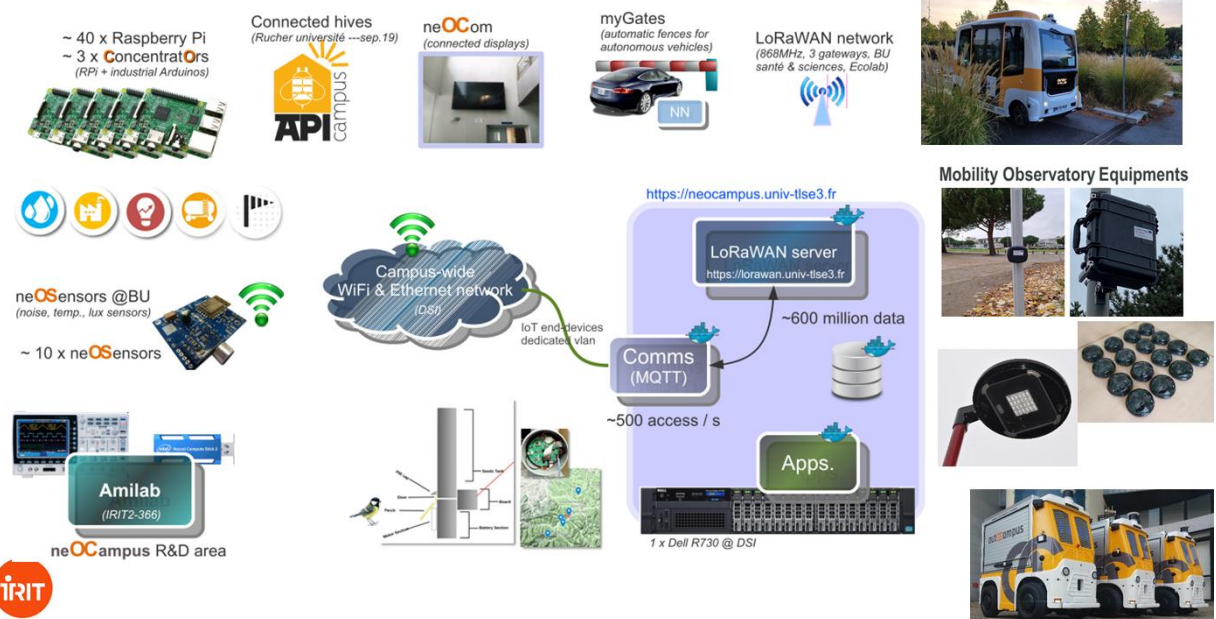
Role : Devices in IoT ecosystem play a crucial role in collecting data, communicating with other devices, and performing specific tasks. Here are some examples of IoT devices:

- Wearable Devices : smartwatches, fitness Trackers, ...
- Smart Home Devices: smart Thermostats, smart lighting systems, home security systems, ...
- Industrial IoT (IIoT) Devices: industrial sensors, connected machines,
- Connected Vehicles: connected cars, fleet Management systems, logistics and maintenance systems,
- Smart Health Devices: remote patient monitoring (vital signs and health conditions), smart Pills, ...
- Agricultural IoT Devices: soil moisture sensors, efficient irrigation systems, livestock Monitoring, ...

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Examples: IoT devices in UT campus



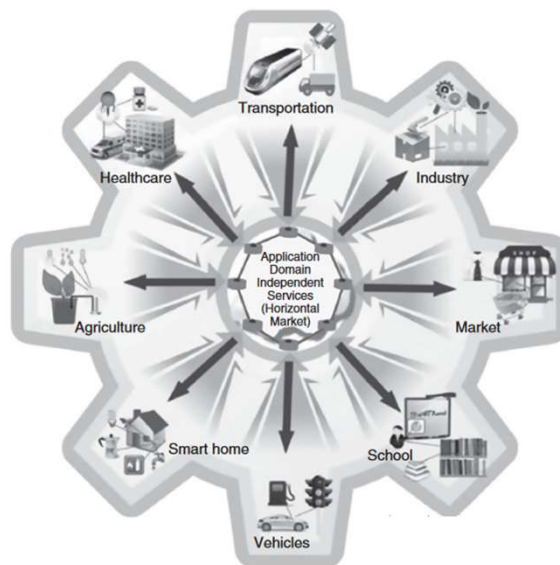
Application Domains

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Application areas

08/09/25



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Application areas



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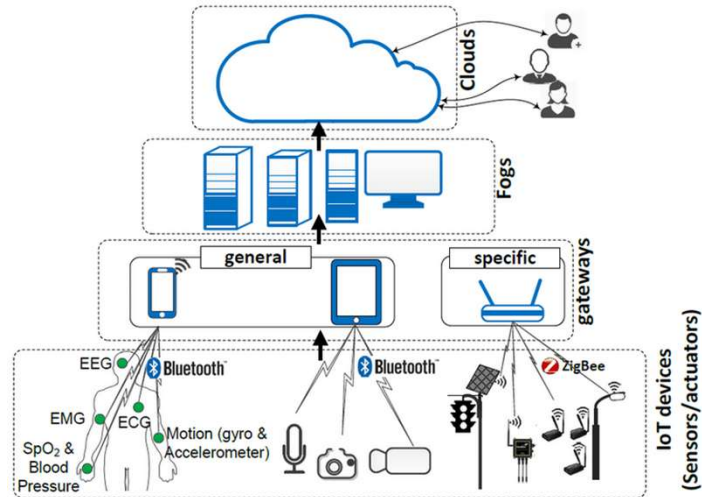
Connectivity

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Rappel - Archi

Les IoT layers (couches)



La vision Business

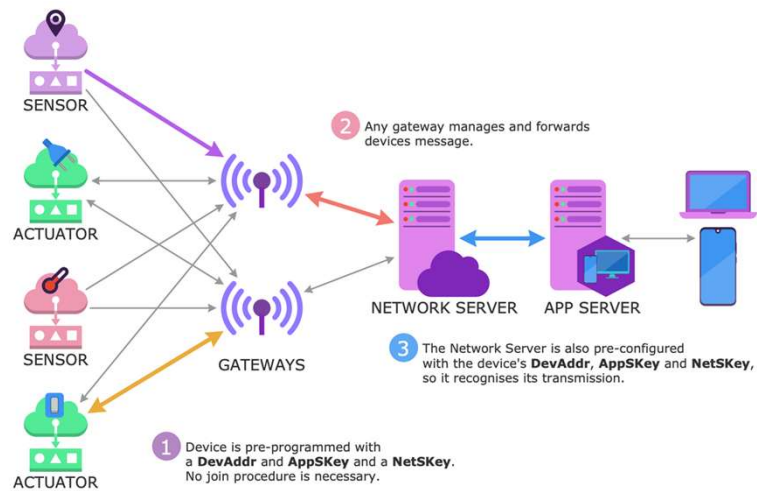


Levels

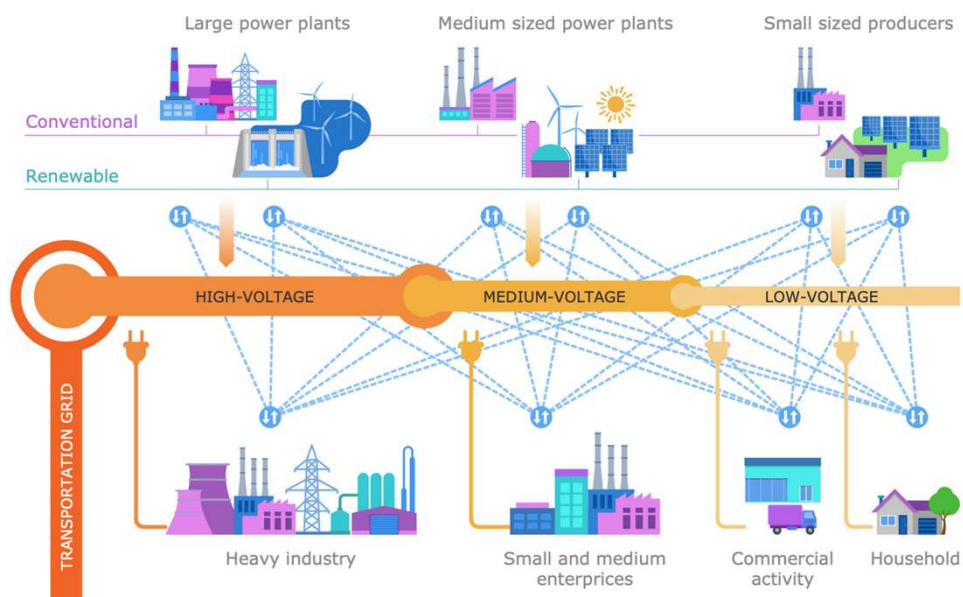
- 7 Collaboration & Processes**
(Involving People & Business Processes)
- 6 Application**
(Reporting, Analytics, Control)
- 5 Data Abstraction**
(Aggregation & Access)
- 4 Data Accumulation**
(Storage)
- 3 Edge Computing**
(Data Element Analysis & Transformation)
- 2 Connectivity**
(Communication & Processing Units)
- 1 Physical Devices & Controllers**
(The "Things" in IoT)



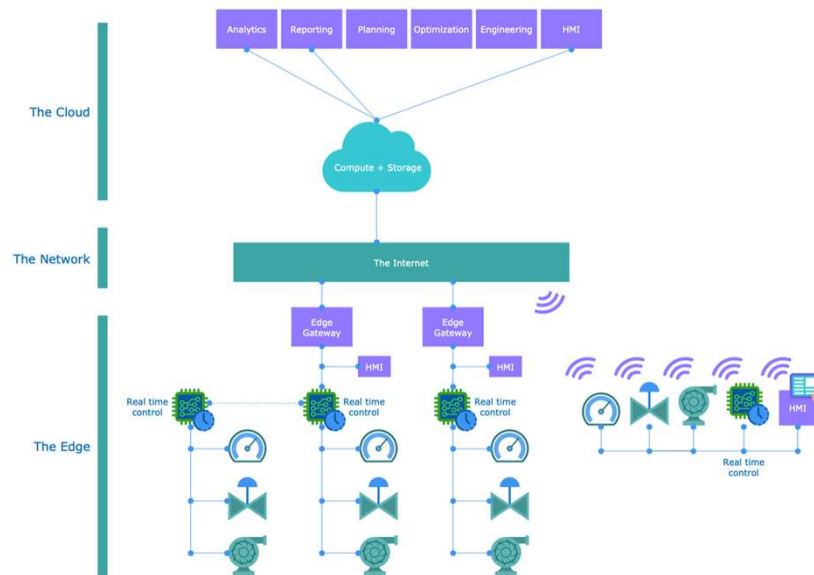
Exemple 1



Exemple 2 - Future Power Grid

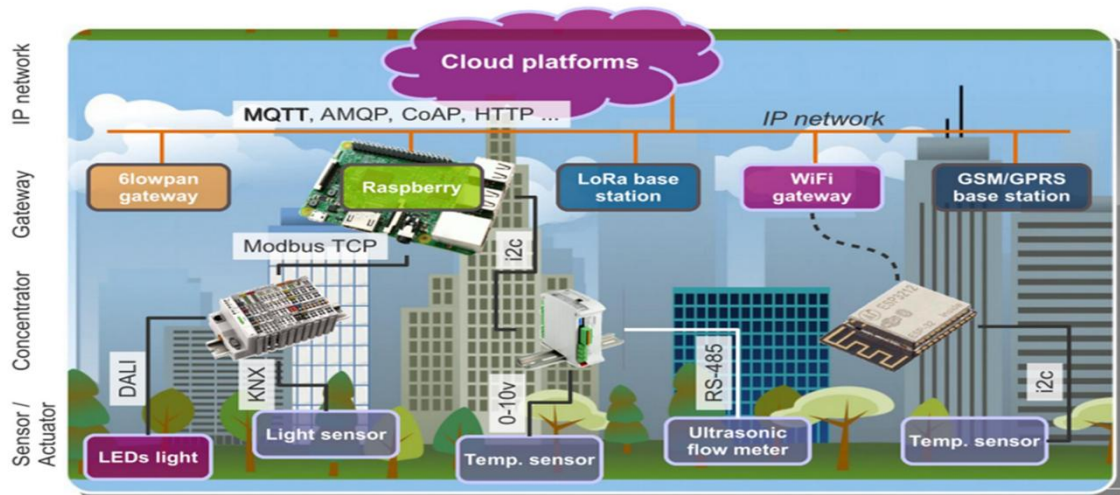


Exemple 3 - IIoT

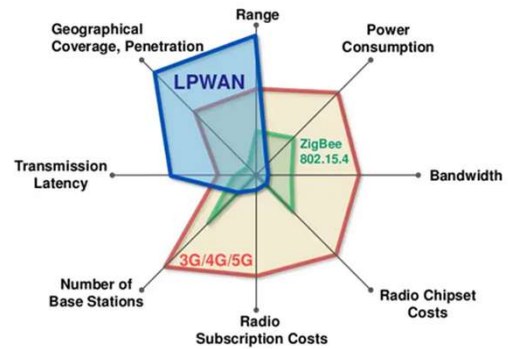
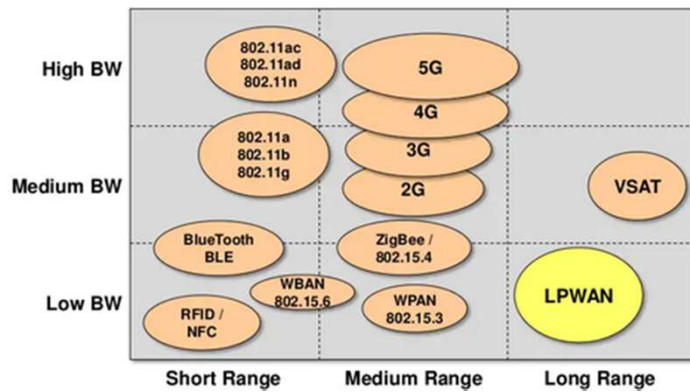


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... and its *Cloudification!*



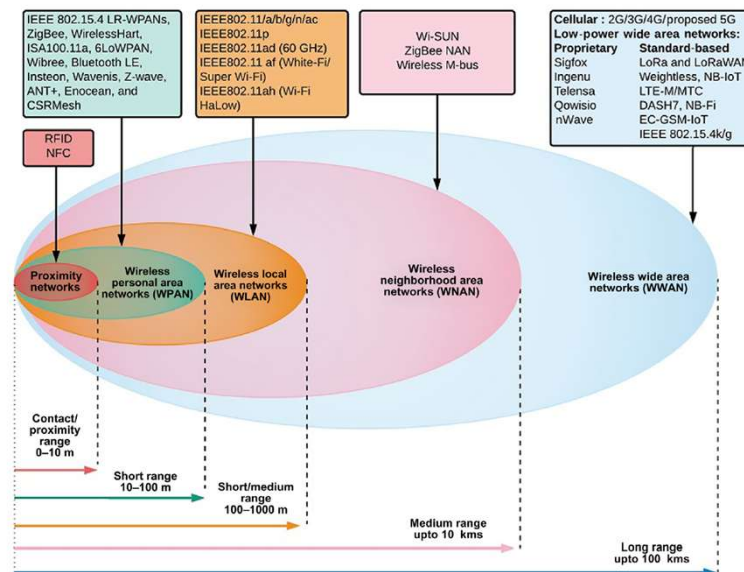
Wireless communication technologies



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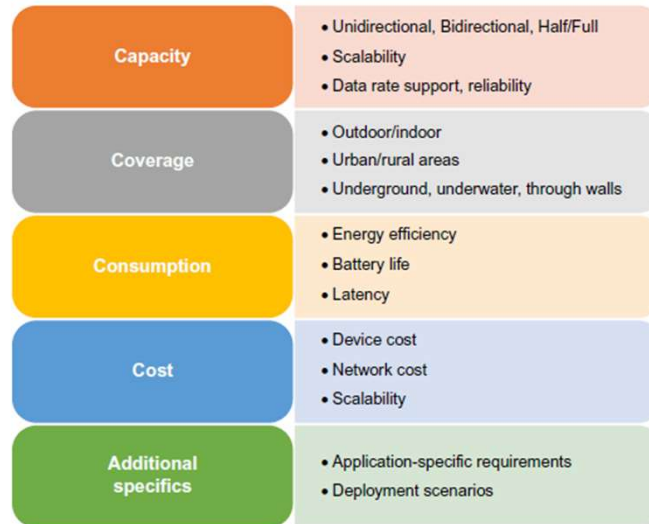
Wireless access geographic coverage



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Application requirement priorities and characteristics



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Factors Influencing Sensor Network Design



- Fault Tolerance (Reliability)
- Scalability
- Production Costs
- Hardware Constraints
- Sensor Network Topology
- Operating Environment (Applications)
- Transmission Media
- Power Consumption (Lifetime)

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Design considerations



- Traffic characteristics
- Capacity and densification
- Energy-efficient operations and low-power sources
- Coverage
- Localization
- Security and privacy
- Reduced device hardware complexity
- Range of solutions options
- Operations, interrelationships, and interworking

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Proprietary and standards-based solutions



❖ **Proprietary technologies**

- ❖ Sigfox
- ❖ Ingenu
- ❖ Telensa
- ❖ Qowiso
- ❖ Nwave

❖ **Standards-based technologies**

- ❖ LoRa and LoRaWAN
- ❖ Weightless
- ❖ Narrowband Internet of things
- ❖ LTE-M
- ❖ DASH7
- ❖ NB-Fi
- ❖ Enhanced coverage global system for mobile Internet of things (EC-GSM-IoT)
- ❖ IEEE 802.15.4k
- ❖ IEEE 802.15.4g

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Proprietary and standards-based solutions

	Short-range network			LPWAN			
	BLE	ZigBee	Wi-Fi	LoRa	Sigfox	NB-IoT	LTE Cat-M1
Frequency	2.4 GHz	Sub-GHz/ 2.4 GHz	2.4/5 GHz	Sub-GHz	Sub-GHz	Licensed GSM/LTE bands	Licensed LTE bands
ISM	Yes	Yes	Yes	Yes	Yes	No	No
Range	100–400 m	100 m	50 m	15 km	50 km	15 km	11 km
Data rate	<25 Mbps	250 kbps	600 Mbps	50 kbps	1 kbps	250 kbps	1 Mbps
Power	Low	Low	High	Low	Low	Low	Low

Note: BLE, bluetooth low energy.

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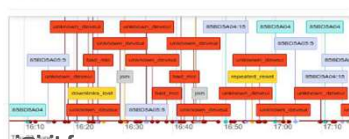
Proprietary and standards-based solutions

11/09/25

	Short-range network			LPWAN			
	BLE	ZigBee	Wi-Fi	LoRa	Sigfox	NB-IoT	LTE Cat-M1
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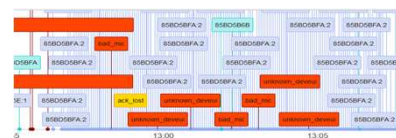
Note: BLE, bluetooth low energy.

Dashboard



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//\ Duty-Cycle
0,1 - 1 %



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Implementation factors



LPWAN technology	Coverage	Alliances and standards	Commercial devices availability
LoRa	Private/some countries of Europe, America, Asia, Oceania, and Africa	LoRa Alliance ETSI LTN	Very high
SigFox	Europe, America, Australia, and areas of Africa and Asia	ETSI LTN	Very high
NB-IoT	Europe, North America, Brazil, Argentina, Russia, China, and Oceania	3GPP	High
LTE-CatM	Europe, North America, Brazil, Argentina, Russia, China, and Oceania	3GPP	High
SNOW	—	—	Very low
Weightless	Private	Weightless Special Interest group	Low
Ingenu-RPMA	United States, China, South Africa, and Italy	Wi-SUN Alliance IEEE 802.15.4k,g	Very low
Telensa	—	ETSI LTN	Low
GSM-IoT	Global (GSM coverage)	EC-GSM-IoT Group 3GPP	Low
Wi-SUN	Private	IEEE 802.15.4k	Low
DASH7	Private	DASH7 Alliance	Moderate
IQRF	Private	IQRF Alliance	Low
MIOTY	Private	—	Low

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More capabilities!



LPWAN technology	Remote firmware updating score (1 = worst, 5 = best)	Location services score (1 = worst, 5 = best)	IP support	Roaming support
LoRa	2	3	Under research	Under research
SigFox	3	5	Under research	Yes
NB-IoT	3	3	Under research	Under research
LTE-CatM	5	3	Yes	Under research
SNOW	1	2	No	—
Weightless	5	2	No	Yes
Ingenu-RPMA	4	2	Yes	No
Telensa	1	1	No	—
GSM-IoT	1	3	No	Under research
Wi-SUN	5	3	No	Yes
DASH7	5	3	No	Yes
IQRF	5	2	No	Yes
MIOTY	3	1	No	—

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LoRaWAN

• LoRa network

- is a **LPWAN** (Low Power Wide Area Network) like SigFox,
- technology acquired by Semtech in 2012,
- LoRa is mostly a layer 1 (OSI stack) ranging from 169MHz to 1GHz,
- Mostly known implementation focus on the 433/868MHz (ISM) frequency,
- Chirp spread spectrum (CSS) modulation (FHSS for ZibBee and BT),
- enables point to multi-points networks,
- LoRa modules available (like RFM95W with a 20dBm PA),
- LoRaWAN is (OSI) uppers layers,
- **Examples**, Orange and others have deployed LoRaWAN by the end of 2017.



[2020] Heltec Cube Cell, LoRaWAN arduino compatible board

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TTN indoor gateway



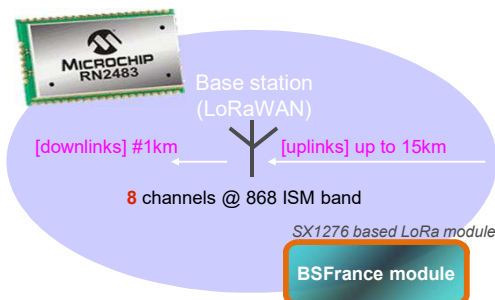
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LoRaWAN

• LoRa network (cont'd)

- Max. 50kbps,
- App. payload up to 222 bytes,
- 125 to 250kHz bandwidth,
- 1% & 10% duty-cycle @ 868MHz,
- LoRaWAN server @ base station,
- LoRaWAN manages data-rate AND power of each end-device, (Adaptive Data Rate –ADR)
- Allow Private Network

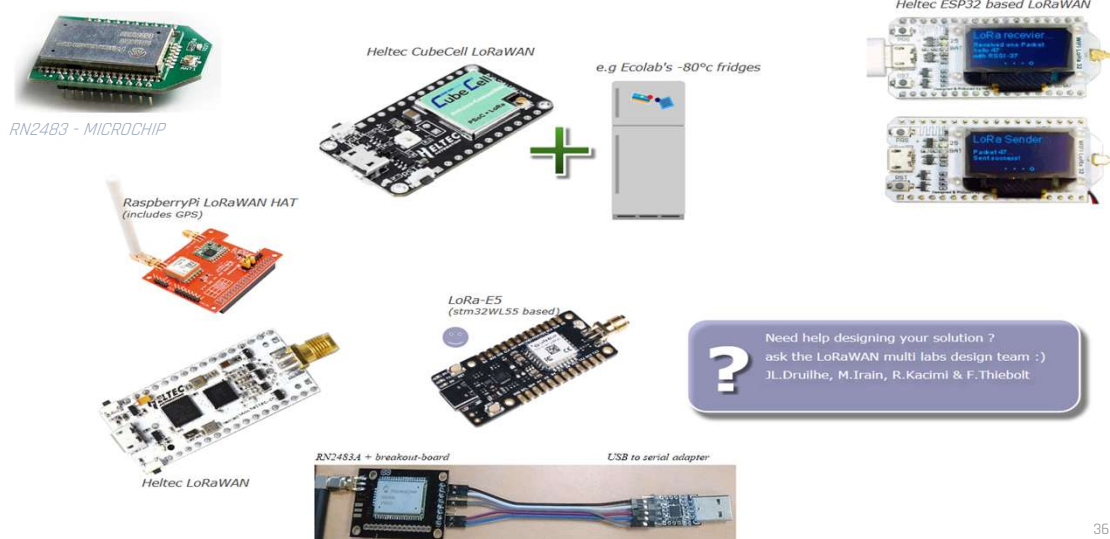
<https://www.lora-alliance.org>



BSFrance LoRaM3-D L151 / STM32 + OLED + LiPo + SX1276
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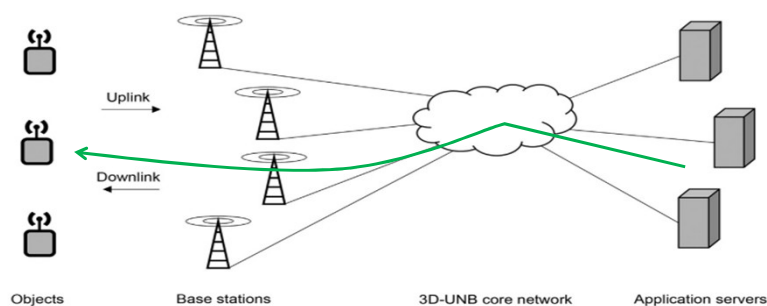
LoRaWAN Devices



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Ateliers Expérimentation et Instrumentation

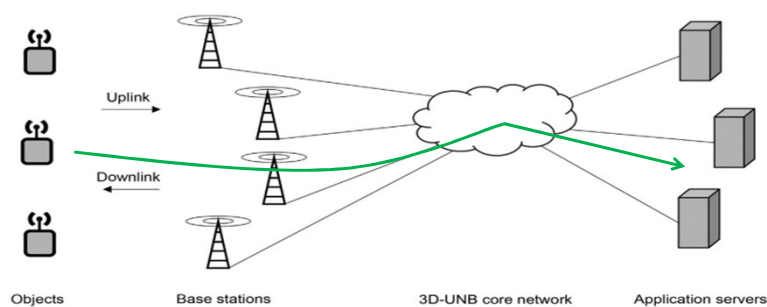
The downlink limitation



Mainly needed for remote configuration and firmware upgrade.

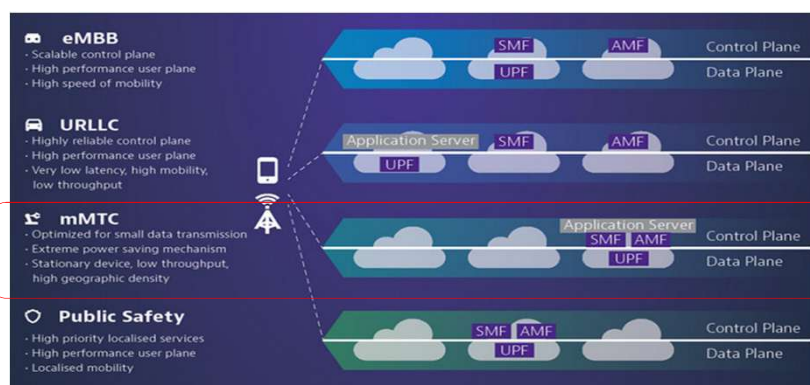
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Ateliers Expérimentation et Instrumentation



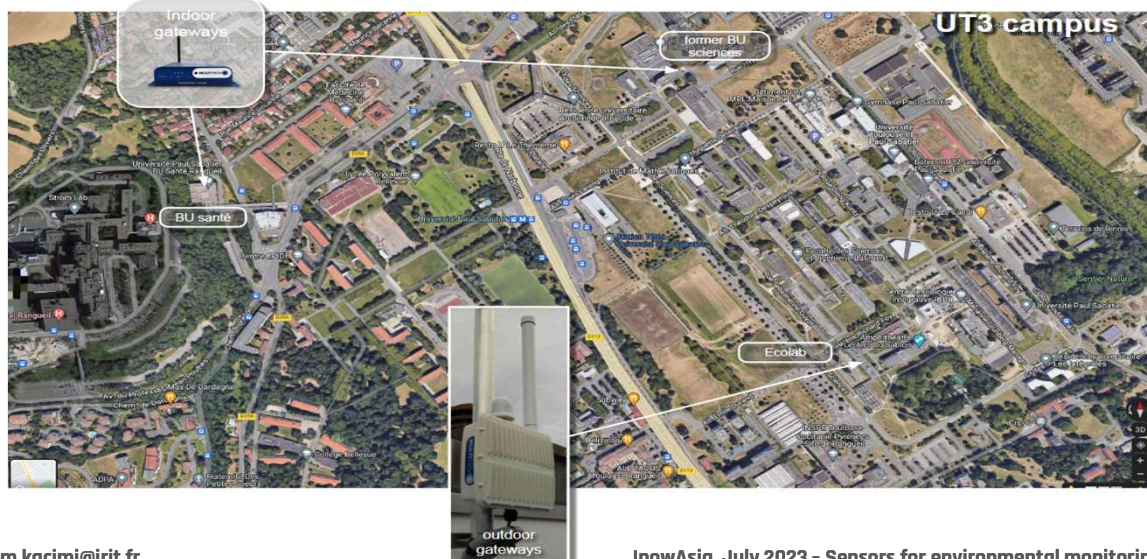
Mainly needed for remote configuration and firmware upgrade.

The future 5G slice on mMTC!



mMTC : massive Machine Type Communications

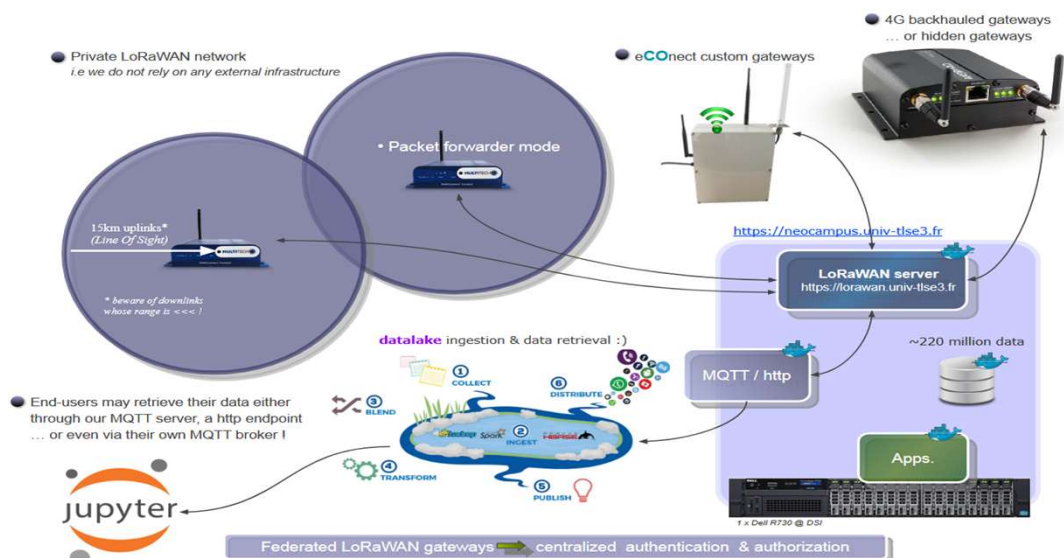
UT3 LoRaWAN Infrastructure



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UT3 LoRaWAN Infrastructure



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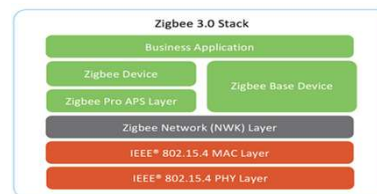
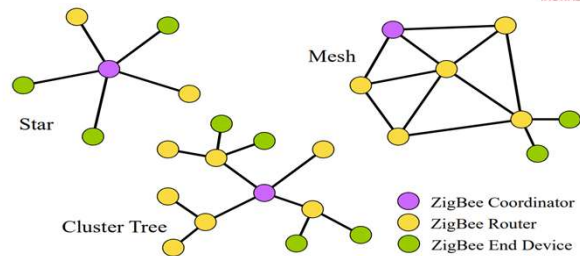
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ZigBee



DIGI XBEE 3 ZIGBEE 3.0

Bande 2,4 GHz
Débit 250 Kbps
Délai 5 ms
Portée ~100-3200 m
Topologie Star, Tree, Mesh



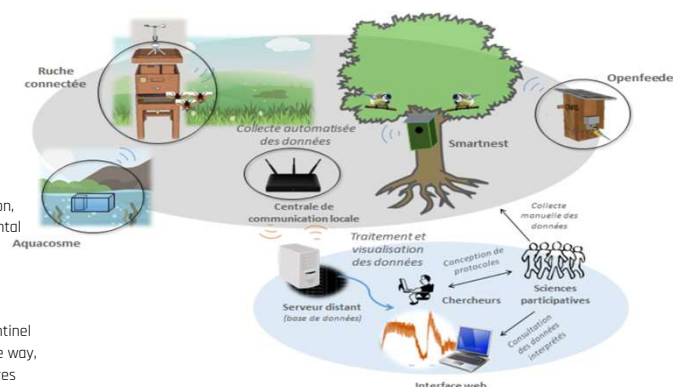
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Example 1: ECOnect project

❖ Data communication and Data processing architecture

- ❖ Develop an architecture for collection, transfer and analysis of environmental data
- ❖ Apply this architecture to three environment sentinel systems
- ❖ Evaluate the relevance of these sentinel systems to assess, in an integrative way, the effect of anthropogenic pressures



"Development of environmental sentinel systems, connected, to better understand the degradation of waterways, the decline of bees and birds"

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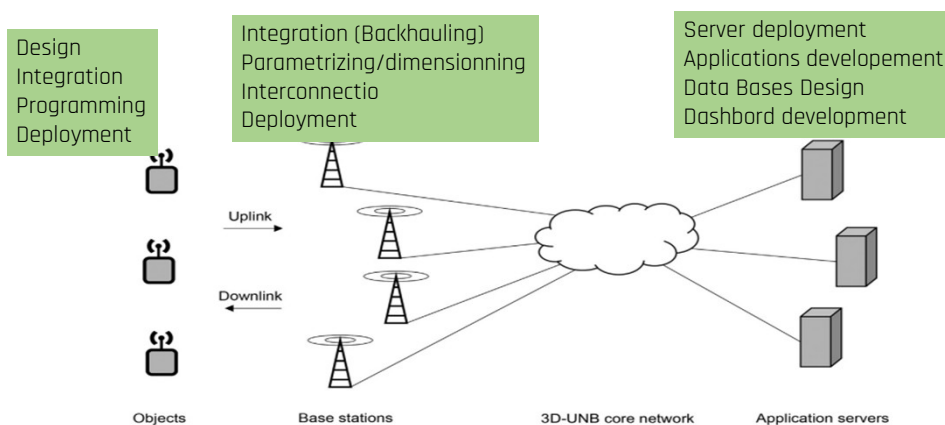
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End-to-End dev

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End-to-End Architecture



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Programming

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Program architecture : example !



```
# Copyright (C) 2023 M. Irain, R. Kacimi
# This file is part of sense-hat-iot-project.
#
# sense-hat-iot-project is free software!
```

```
from devices import Microchip
from formats import NeoCayenneLPP
from sense_hat import SenseHat
from sys import getsizeof
from time import sleep
```

```
import json
```

```
SENSE_HAT_CHANNEL = 1
```

```
DEVEUI = 'DEADDEAD00090001'
APPEUI = 'DEAD25DEAD25DEAD'
APPKEY = '0123456789ABCDEF0123456789ABCDEF'
```

```
def print_data(data):
    print('Mesures: ')
    if isinstance(data, bytes):
        data = { key : value for key, value in map(lambda x
            json.loads(NeoCayenneLPP.from_bytes(data).to_json())
        for s in data:
            print(f' {s}: {data[s]}')
```

```
def print_sizes(data, formatted):
    print(f'Raw size: {getsizeof(json.dumps(data))} bytes; F
```

```
def get_data_from_sense_hat(sh):
    data = {}
    data['temperature'] = sh.get_temperature()
    data['humidity'] = sh.get_humidity()
    data['pressure'] = sh.get_pressure()
    return data
```

```
def format_data_1(data):
    return json.dumps(data)
```

```
def main():
    sh = SenseHat()
    sh.clear()

    lora_device = Microchip(deveui, appeui, appkey)

    while True:
        data = get_data_from_sense_hat(sh)
        print_data(data)

        formatted = format_data_2(data)
        print_data(formatted)

        print_sizes(data, formatted)

        lora_device.send(formatted)

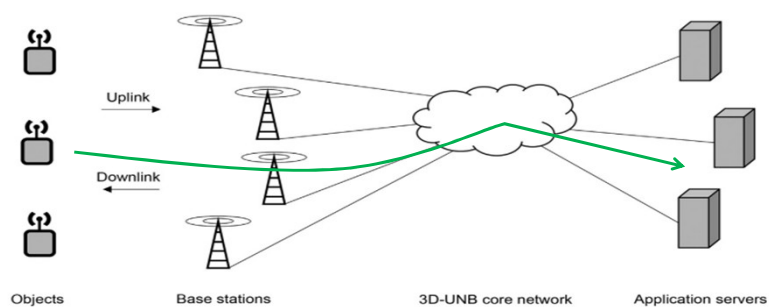
        sleep(10)

if __name__ == '__main__':
    main()
```

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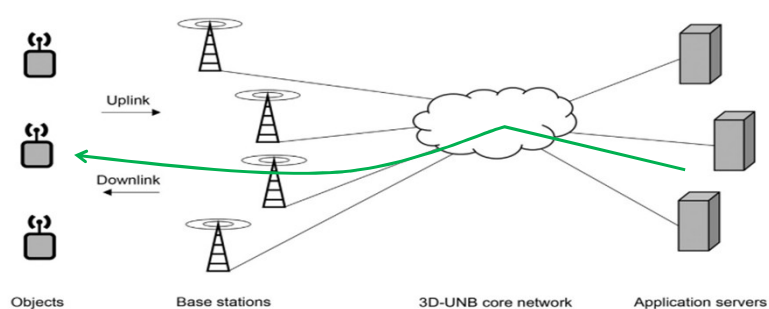
Data Collection (UP-Link)



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Remote Control (Update firmware, ...) (Down-Link)



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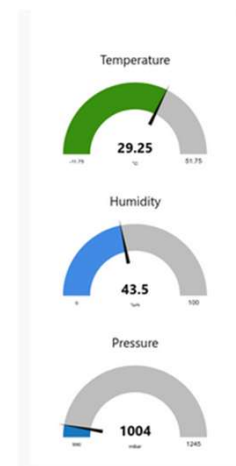
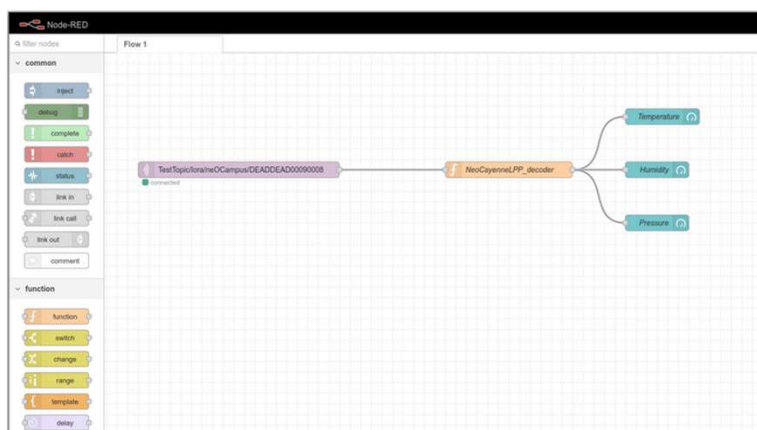


Node-Red

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IoT programming : node-red



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Thank you !

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